

THE KING'S SCHOOL

2010
Higher School Certificate Course
Trial Examination

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks – 120

- Attempt Questions 1-10
- All questions are of equal value

Disclaimer

This is a Trial Higher School Certificate Examination only. Whilst it reflects and mirrors both the format and topics of the Higher School Certificate Examination designed by the NSW Board of Studies for the respective sections, there is no guarantee that the content of this examination exactly replicates the actual Higher School Certificate Examination.

Total marks – 120
Attempt Questions 1-10
All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) Find, correct to the nearest integer, $3420 \sin \frac{\pi}{5}$ **2**
- (b) Find the sum of the first 30 terms of the arithmetic series $9 + 13 + 17 + \dots$ **2**
- (c) Write down a primitive function of $\sec^2(2x + 1)$ **2**
- (d) Simplify $2\sqrt{3} + \frac{2}{\sqrt{3} - 1}$ **2**
- (e) Let $f(x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots - \frac{x^{2010}}{2010}$
Find the value of $f'(-1)$ **2**
- (f) Sketch the hyperbola $y - 1 = \frac{1}{x}$, clearly showing its asymptotes and any intercepts made with the coordinate axes. **2**

End of Question 1

(a) State the domain of the function $f(x) = \ln(1 + x)$ 1

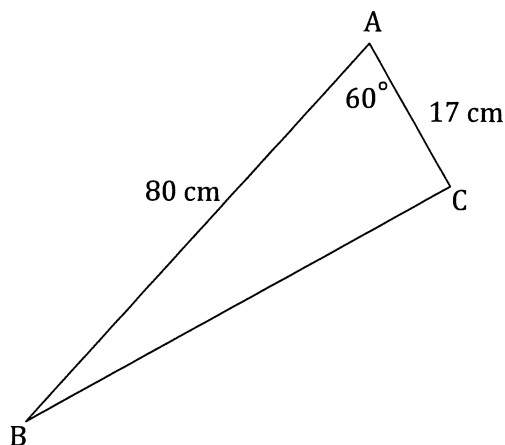
(b) Differentiate

(i) $\sin(x^2)$ 2

(ii) $x^2 \ln x$ 2

(c) Evaluate $\int_{-2}^2 \frac{2}{\sqrt{2x+5}} dx$ 3

(d)

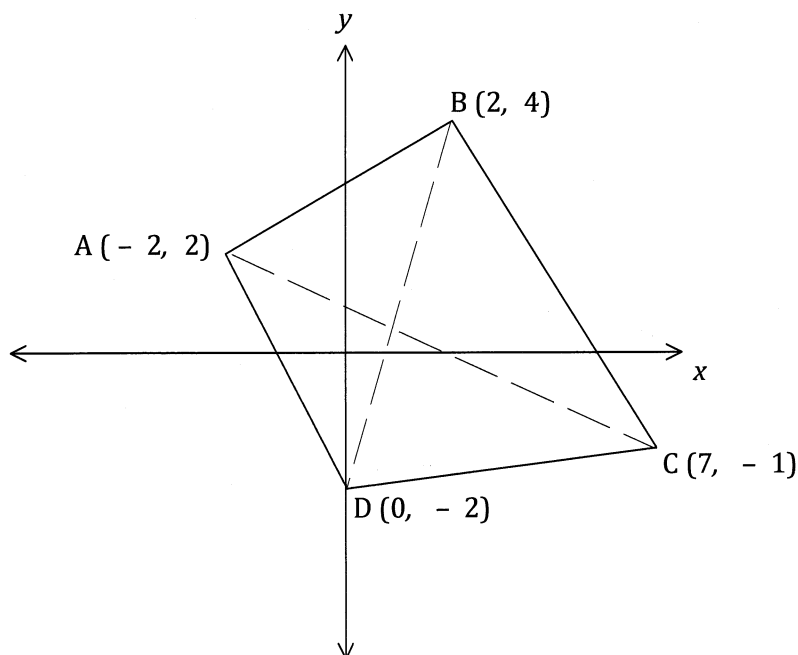


(i) Find the exact area of $\triangle ABC$ 2

(ii) Find the length of BC 2

End of Question 2

(a)



The diagram shows the quadrilateral ABCD

- | | |
|--|---|
| (i) Show that the diagonals AC and BD are perpendicular. | 2 |
| (ii) Find the mid-point M of BD. | 1 |
| (iii) Show that M lies on the diagonal AC. | 2 |
| (iv) Hence, or otherwise, find the area of the quadrilateral ABCD. | 2 |

- (b) (i) On the same axes, sketch the parallel lines

$$3x + 4y = 12 \quad \text{and} \\ 3x + 4y = 24,$$

clearly showing their intercepts made with the coordinate axes. 2

- | | |
|---|---|
| (ii) Find the perpendicular distance between the lines. | 2 |
| (iii) Hence, or otherwise, find the area of the region enclosed by the two lines in the first quadrant. | 1 |

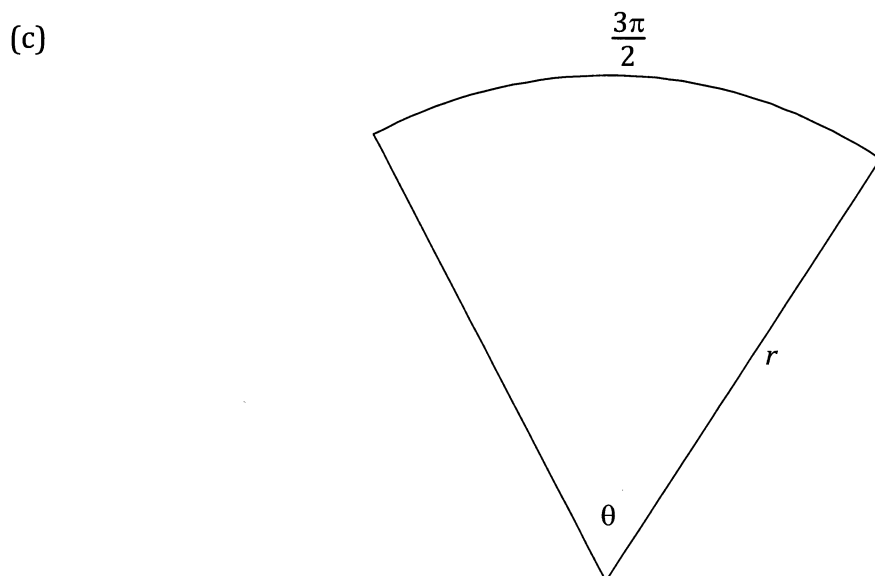
End of Question 3

(a) (i) Prove that $\sin x + \cot x \cos x \equiv \operatorname{cosec} x$ **3**

(ii) Find the exact solutions of $\sin x + \cot x \cos x = 2$, $0 < x < 2\pi$ **2**

(b) $S = \frac{\pi^4}{13} + \frac{\pi^3}{13} + \frac{\pi^2}{13} + \dots$ is a geometric series.

Show that $S < 11$ **3**



The diagram shows a sector of a circle with radius r and central angle θ radians.

The arc length is $\frac{3\pi}{2}$ and the area of the sector is 12π

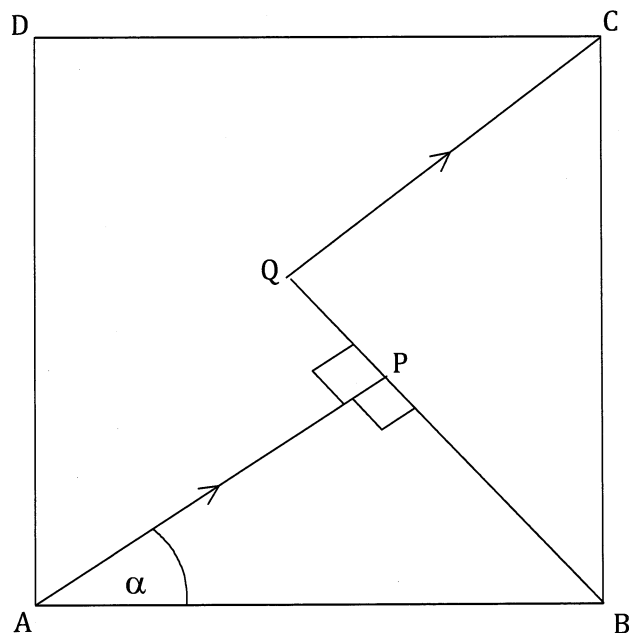
(i) Show that $r^2\theta = 24\pi$ **1**

(ii) Find the value of r **2**

(iii) Find the central angle correct to the nearest degree. **1**

End of Question 4

(a)



ABCD is a square. P is a point interior to the square so that $\angle APB = 90^\circ$.

BP is produced to Q so that $AP \parallel QC$

Let $\angle PAB = \alpha$

(i) Explain why $\angle CQB = 90^\circ$ 1

(ii) Prove that $\triangle APB \equiv \triangle CQB$ 3

(iii) Deduce that $PQ = AP - PB$ 2

(b) Let $Q(x) = (k + 1)x^2 - 2(k + 1)x + k$

Prove that there are no values of k for which $Q(x)$ is positive definite. 3

(c) Solve

(i) $x^2 > 2$ 1

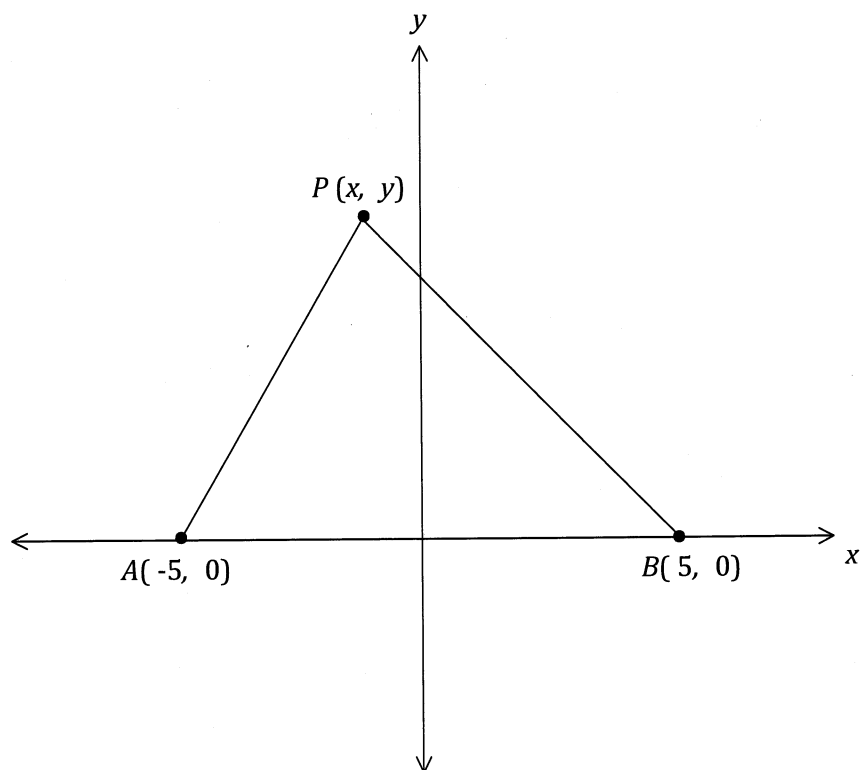
(ii) $|x^2 - 9| < 7$ 2

End of Question 5

-
- (a) (i) Find the sum and product of the roots of the equation $3x^2 + 6x - 11 = 0$ **2**
- (ii) The line $y = 3x + 6$ meets the hyperbola $y = \frac{11}{x}$ at the points P and Q .
Find the coordinates of the mid-point M of the interval PQ . **3**
- (b) A particle moves on the x axis with its velocity v at any time $t \geq 0$ given by
 $v = e^t - 12e^{-t}$
The particle is initially at the origin.
- (i) Show that the acceleration is positive at all times. **2**
- (ii) The particle stops when $e^t = \sqrt{12}$. **[DO NOT SHOW THIS]**
Find the exact acceleration when the particle stops. **1**
- (iii) Find the displacement x at any time t . **2**
- (iv) Find the time taken for the particle to return to the origin. **2**

End of Question 6

(a)



$A(-5, 0)$ and $B(5, 0)$ are two fixed points. $P(x, y)$ is any point so that $\frac{PB}{PA} = \frac{3}{2}$

(i) Show that $(-1, 0)$ is a point on the locus of $P(x, y)$ **1**

(ii) Prove that the equation of the locus of $P(x, y)$ is

$$x^2 + y^2 + 26x + 25 = 0 \quad \text{3}$$

(iii) Describe the locus of P in precise geometrical terms. **2**

(b) Let $f(x) = 1 - \cos \pi x$, $-1 \leq x \leq 1$

(i) State the period of the function. **1**

(ii) Sketch the graph of $y = f(x)$ **2**

(iii) Find the area bounded by the curve and the x axis, $-1 \leq x \leq 1$ **3**

End of Question 7

- (a) Use Simpson's Rule with three function values to approximate

$$\int_0^1 \sqrt{18x^2 - 3x + 1} \, dx$$

2

- (b) Let $f(x) = x - 2\sqrt{x} - 3$

- (i) By letting $u = \sqrt{x}$, or otherwise, solve $f(x) = 0$

2

- (ii) Find any stationary points and determine their nature.

3

- (iii) Sketch the graph of $y = f(x)$ indicating the x and y intercepts, if any.

2

- (iv) The line $y = x - 4$ is a normal to the curve $y = f(x)$ at the point P .

Find the coordinates of P .

3

End of Question 8

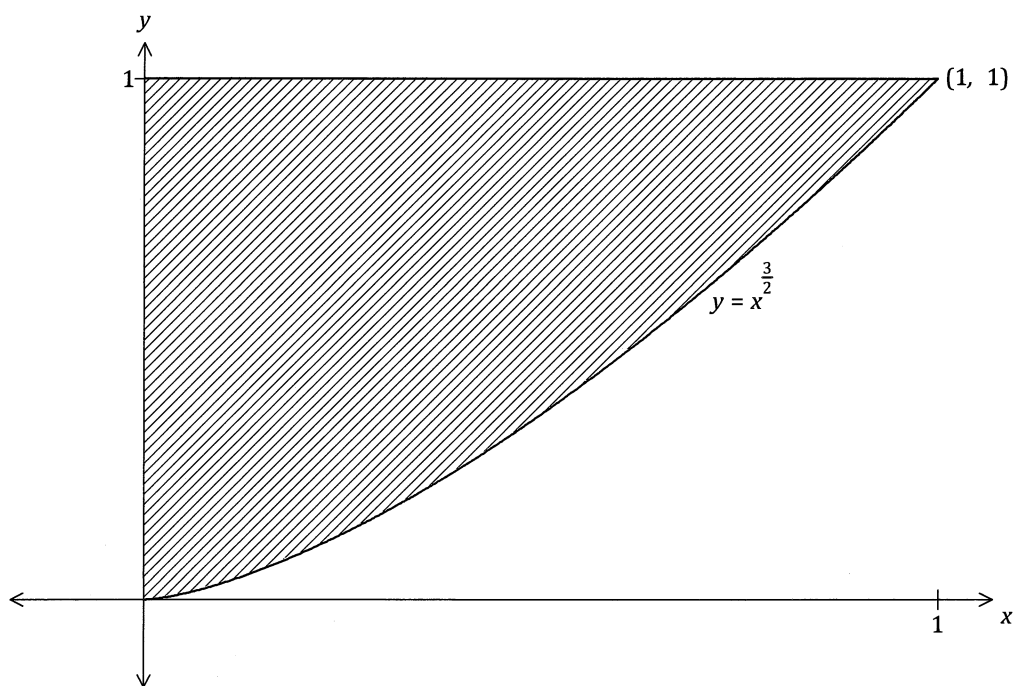
- (a) A balloon is being inflated at a rate $R \text{ cm}^3/\text{s}$ given by $R = \frac{100t}{1+t^2} + 10$,
where $t \geq 0$ is the time in seconds.

(i) At what rate is the balloon being inflated after 2 seconds? **1**

(ii) Determine the greatest rate of inflation. **3**

(iii) The balloon held 69 cm^3 of air initially. Find, correct to 3 significant figures, the volume of air held by the balloon after 10 seconds. **3**

(b) (i)

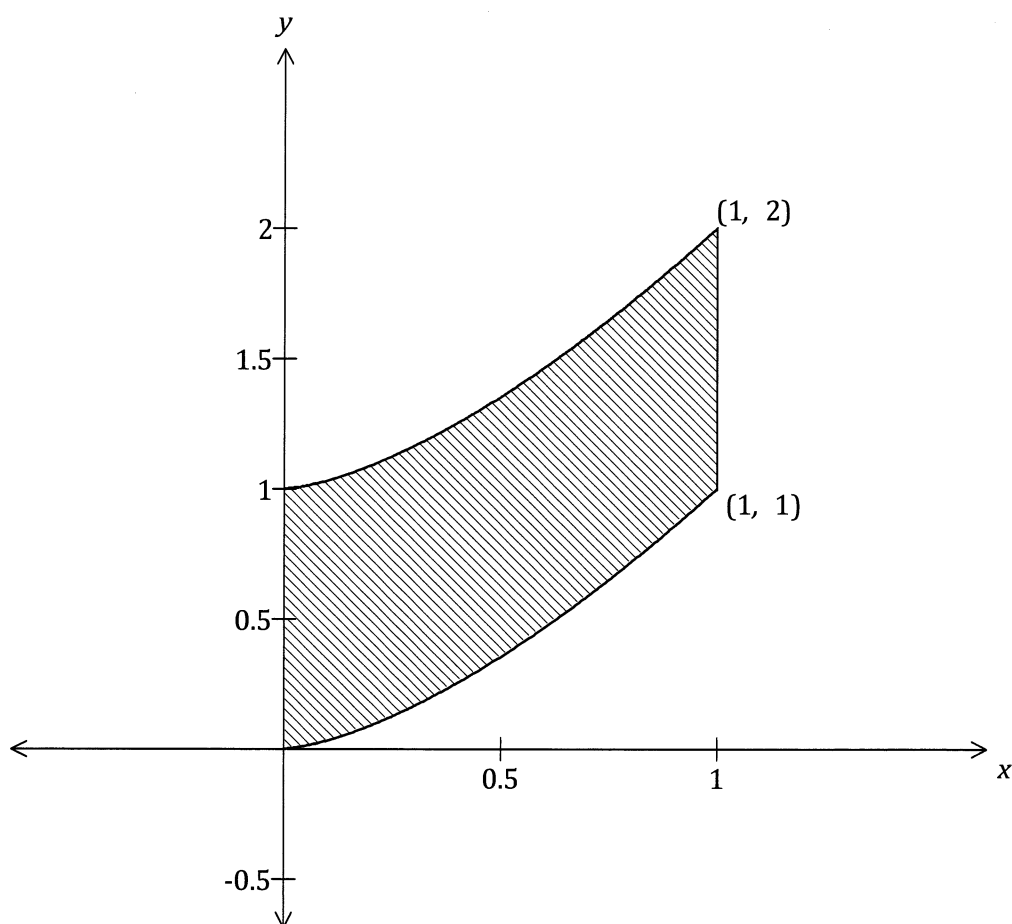


The diagram shows the shaded region between the curve $y = x^{\frac{3}{2}}$, the y axis and the line $y = 1$. This region is revolved about the y axis.

Find the volume of the solid generated. **3**

Question 9 continues on the next page

(b) (ii)



The diagram shows the shaded region between the curves $y = x^{\frac{3}{2}}$ and $y = x^{\frac{3}{2}} + 1$, the y axis and the line $x = 1$.

This region is revolved about the y axis.

Find the volume of the solid generated.

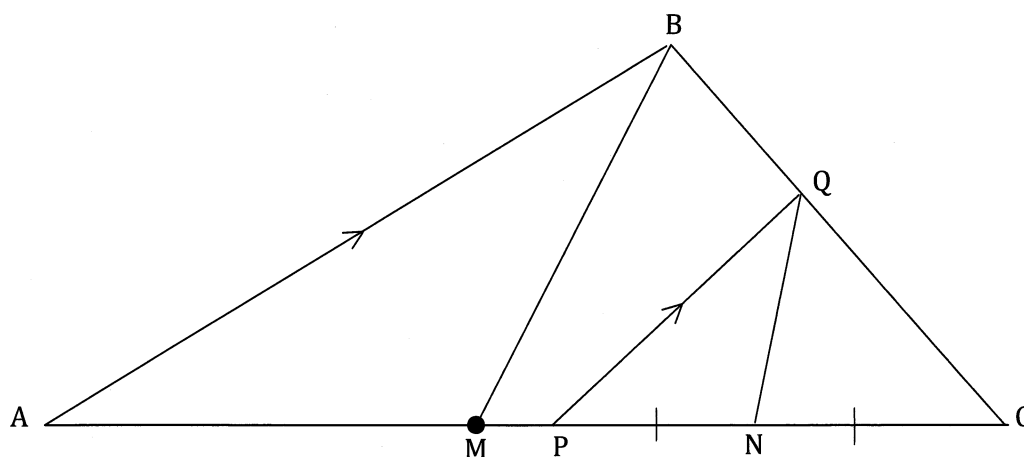
2**End of Question 9**

(a) Find the smallest integer n for which $0.5^n < 12 \times 10^{-2010}$ 2

(b) (i) Sketch the graph of the function $y = 2|x| - x + 1$ 2

(ii) Evaluate $\int_{-1}^1 2|x| - x + 1 \, dx$ 2

(c)



In the diagram, M is the mid-point of side AC of $\triangle ABC$.

P is on AC and Q is on BC so that $PQ \parallel AB$.

N is the mid-point of PC.

(i) Explain why $\frac{CB}{CQ} = \frac{CA}{CP}$ 1

(ii) Deduce that $\frac{CB}{CQ} = \frac{CM}{CN}$ 1

(iii) Explain why $\triangle CBM$ is similar to $\triangle CQN$ 2

(iv) Prove that $MB \parallel NQ$ 2

End of Examination Paper

Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

Note: $\ln x = \log_e x, \quad x > 0$

Question 1

(a) 2010

(b) $S_{30} = \frac{30}{2} (18 + 29 \times 4) = 2010$

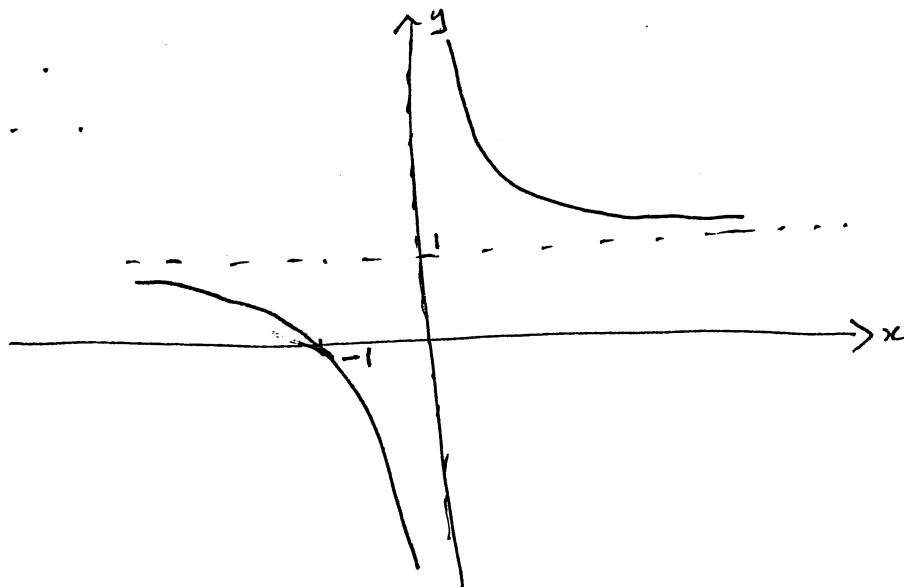
(c) $\frac{1}{2} \tan(2x+1)$

(d) $2\sqrt{3} + \frac{2(\sqrt{3}+1)}{(\sqrt{3}-1)(\sqrt{3}+1)} = 2\sqrt{3} + \frac{2(\sqrt{3}+1)}{2} = 3\sqrt{3} + 1$

(e) $f'(x) = 1 - x + x^2 - x^3 + \dots - x^{2009}$

$\therefore f'(-1) = 1 + 1 + 1 + \dots + 1$, 2010 terms
 $= 2010$

(f) $x \neq 0, y \neq 1$; $y=0, x=-1$



Question 2

(a) $1+x > 0 \Rightarrow$ domain is $x > -1$

(b) (i) $2x \cos(x^2)$

(ii) $x^2 \cdot \frac{1}{x} + \ln x \cdot 2x$ (will do)

$$= x + 2x \ln x \quad \text{or} \quad x(1 + 2 \ln x)$$

(c) $I = \int_{-2}^2 2(2x+5)^{-\frac{1}{2}} dx$

$$= 2 \left[\frac{(2x+5)^{\frac{1}{2}}}{\frac{2}{2}} \right]_{-2}^2 = 2 \left[\sqrt{2x+5} \right]_{-2}^2 = 2(3-1) = 4$$

(d) (i) $A = \frac{1}{2} \cdot 80 \cdot 17 \sin 60^\circ \text{ cm}^2$

$$= 40 \cdot 17 \cdot \frac{\sqrt{3}}{2} \text{ cm}^2 = 340\sqrt{3} \text{ cm}^2$$

(ii) $BC^2 = 80^2 + 17^2 - 2 \cdot 80 \cdot 17 \cos 60^\circ$

$$= 80^2 + 17^2 - 80 \times 17 = 5329$$

$$\therefore BC = \sqrt{5329} \text{ cm} = 73 \text{ cm}$$

Question 3

$$(a) (i) \text{ grad } AC = \frac{-1-2}{7+2} = -\frac{3}{9} = -\frac{1}{3}$$

$$\text{grad } BD = \frac{-2-4}{0-2} = 3$$

$$\text{+ since } -\frac{1}{3} \times 3 = -1 \text{ then } AC \perp BD$$

$$(ii) M = (1, 1)$$

$$(iii) \text{ grad } MA = \frac{2-1}{-2-1} = -\frac{1}{3} = \text{grad } AC$$

$\Rightarrow A, M, C$ collinear i.e. M is on AC

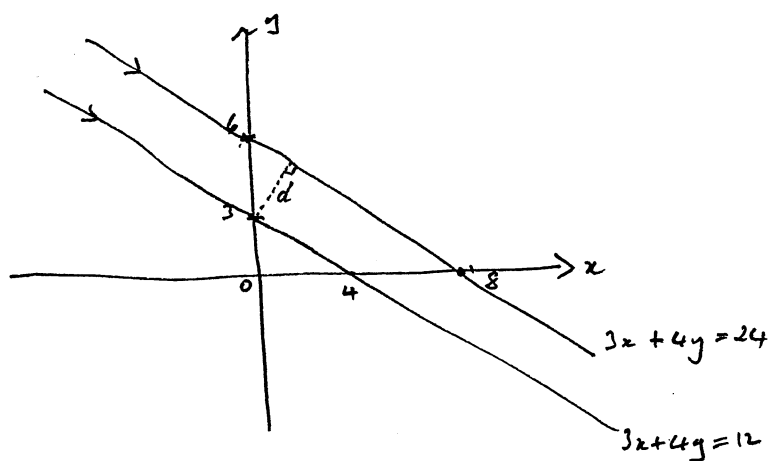
$$(iv) \text{ Area} = \frac{1}{2} \cdot AC \cdot BM + \frac{1}{2} \cdot AC \cdot DM$$

$$= \frac{1}{2} AC (BM + DM) = \frac{1}{2} \cdot AC \cdot BD$$

$$= \frac{1}{2} \cdot \sqrt{9^2 + 3^2} \cdot \sqrt{2^2 + 6^2}$$

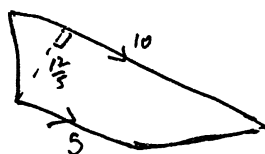
$$= 30$$

(b) (i)



$$(ii) d = \frac{|0 + 12 - 24|}{\sqrt{3^2 + 4^2}} = \frac{12}{5}$$

(iii)



TRAPEZIUM

$$A = \frac{1}{2} (10 + 5) \frac{12}{5} = 18$$

$$\text{OR } A = \frac{1}{2} \cdot 8 \cdot 6 - \frac{1}{2} \cdot 3 \cdot 4 = 18$$

Question 4

$$(a) (i) \quad LS = \sin x + \frac{\cos x}{\sin x} \cdot \cos x$$

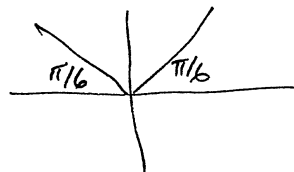
$$= \frac{\sin^2 x + \cos^2 x}{\sin x}$$

$$= \frac{1}{\sin x} = \operatorname{cosec} x$$

$$(ii) \quad \text{From (i), } \operatorname{cosec} x = 2$$

$$\Rightarrow \sin x = \frac{1}{2}$$

$$\therefore x = \frac{\pi}{6}, \frac{5\pi}{6}$$



$$(b) \quad G.S., \quad a = \frac{\pi^4}{13}, \quad r = \frac{1}{\pi} \quad (\& \ -1 < r < 1)$$

$$\therefore S = \frac{\frac{\pi^4}{13}}{1 - \frac{1}{\pi}} = \frac{\pi^5}{13(\pi-1)} = 10.99\dots < 11$$

$\Rightarrow S < 11$ since it's a series of positive terms

$$(c) (i) \quad A = \frac{1}{2} r^2 \theta = 12\pi \Rightarrow r^2 \theta = 24\pi$$

$$(ii) \quad r^2 \theta = 24\pi$$

$$r \theta = \frac{3\pi}{2}$$

$$\therefore r = \frac{24\pi}{\frac{3\pi}{2}} = 16$$

$$(iii) \quad \theta = \frac{3\pi}{2 \times 16} = \frac{3 \times 180}{32}^\circ = 17^\circ, \text{ nearest degree}$$

Question 5

(a) (i) $\angle CQB = \angle QPA = 90^\circ$, alternate $\angle$ s in $\parallel$ lines AP, QC

(ii) $\angle PBA = 90^\circ - \angle^\circ$, $\angle$ sum ΔAPB

$\therefore \angle CBQ = \angle^\circ$, $\angle ABC = 90^\circ$, $ABCD$ a square

$\therefore \sim \Delta$ s APB, CQB

$\angle P = \angle Q = 90^\circ$, (i)

$\angle PAB = \angle CBQ = \angle^\circ$

$AB = BC$, sides of square $ABCD$

$\therefore$ congruent, AAS

(iii) $PQ = QB - PB$

$= AP - PB$ since $QB = AP$, corr. sides in
cong Δ s (ii)

(b) Firstly, $k+1 > 0 \Rightarrow \underline{k > -1}$

Next, $\Delta = 4(k+1)^2 - 4k(k+1)$

$= 4(k+1)(k+1-k)$

$= 4(k+1) < 0$ for P.D.

$\Rightarrow \underline{k < -1}$

$\therefore$ no value of k makes $Q(u)$ P.D.

(c) (i) $x > \sqrt{2}$ or $x < -\sqrt{2}$

(ii) $\therefore -7 < x^2 - 9 < 7$

or $2 < x^2 < 16 \Rightarrow x > \sqrt{2}$ or $x < -\sqrt{2}$

and $-4 < x < 4$

$\therefore -4 < x < -\sqrt{2}$ or $\sqrt{2} < x < 4$

Question 6

$$(a) (i) \Sigma = -\frac{6}{3} = -2, \quad \Pi = -\frac{11}{3}$$

$$(ii) \text{ at } P, Q \quad 3x + 6 = \frac{11}{x}$$

$$\Rightarrow 3x^2 + 6x = 11$$

$$\text{or } 3x^2 + 6x - 11 = 0 \text{ has roots } x_1, x_2 \text{ say}$$

$$\Rightarrow x_1 + x_2 = -2 \Rightarrow \frac{x_1 + x_2}{2} = -1$$

$$\text{i.e. at } M, x = -1, y = -3 + 6 \quad \therefore M = (-1, 3)$$

$$(b) (i) \ddot{x} = e^t + 12e^{-t} > 0 \quad \forall t \text{ since } e^t, e^{-t} > 0 \quad \forall t$$

$$\therefore \ddot{x} > 0 \quad \forall t$$

$$(ii) \ddot{x} = \sqrt{12} + \frac{12}{\sqrt{12}} = \sqrt{12} + \sqrt{12} = 2\sqrt{12} \quad \text{or } 4\sqrt{3}$$

$$(iii) x = e^t + 12e^{-t} + c$$

$$\therefore 0 = 1 + 12 + c, \quad c = -13$$

$$\therefore x = e^t + 12e^{-t} - 13$$

$$(iv) x = 0 \Rightarrow e^{2t} + 12 - 13e^t = 0$$

$$e^{2t} - 13e^t + 12 = 0$$

$$(e^t - 1)(e^t - 12) = 0$$

$$\Rightarrow e^t = 1 \text{ or } 12 \Rightarrow \text{returns to origin when } e^t = 12$$

$$\therefore t = \ln 12$$

Question 7

(a) (i) If $P = (-1, 0)$, $\frac{PB}{PA} = \frac{6}{4} = \frac{3}{2} \therefore \text{result}$

(ii) $3PA = 2PB \text{ or } 9PA^2 = 4PB^2$

$$\therefore 9((x+5)^2 + y^2) = 4((x-5)^2 + y^2)$$

$$\text{i.e. } 9(x^2 + 10x + 25 + y^2) = 4(x^2 - 10x + 25 + y^2)$$

$$\text{i.e. } 5x^2 + 5y^2 + 130x + 125 = 0$$

$$\text{i.e. } x^2 + y^2 + 26x + 25 = 0$$

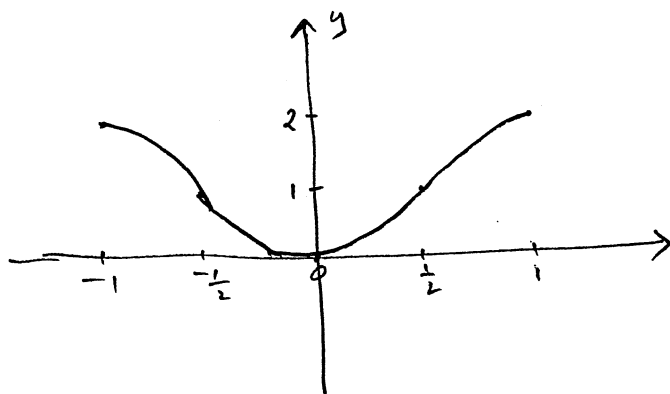
(iii) $\therefore (x+13)^2 - 169 + y^2 = -25$

$$\text{i.e. } (x+13)^2 + y^2 = 144$$

$\Rightarrow$ circle, centre $(-13, 0)$, radius 12

(b) (i) $p = \frac{2\pi}{\pi} = 2$

(ii)



(iii) $A = 2 \int_0^1 1 - \cos \pi x \, dx$

$$= 2 \left[x - \frac{\sin \pi x}{\pi} \right]_0^1$$

$$= 2(1 - 0 - (0)) = 2$$

Question 8

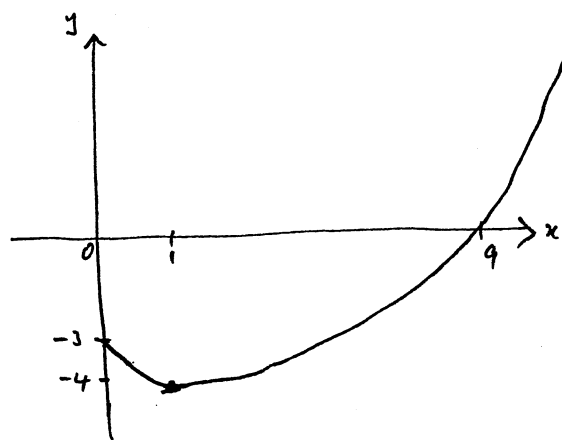
$$(a) \quad I \approx \frac{1}{6} \cdot 1 [1 + 4 + 4 \times 2] \\ = \frac{13}{6}$$

$$(b) \quad (i) \quad f(x) = 0 \Rightarrow (\sqrt{x} + 1)(\sqrt{x} - 3) = 0 \\ \Rightarrow \sqrt{x} = 3 \text{ only as } \sqrt{x} + 1 \neq 0 \\ \therefore x = 9$$

$$(ii) \quad f'(x) = 1 - 2 \cdot \frac{1}{2} x^{-\frac{1}{2}} = 1 - \frac{1}{\sqrt{x}} \\ = 0 \text{ if } x = 1 \\ f''(x) = \frac{1}{2} x^{-\frac{3}{2}} > 0 \text{ if } x = 1 \Rightarrow \cup$$

∴ minimum T.P (1, -4)

$$(iii) \quad x=0, y=-3 \quad ; \quad x=9, y=0, (i)$$



$$(iv) \quad y = x - 4 \text{ has gradient } 1 \Rightarrow \text{gradient of tangent} = -1$$

$$f'(x) = 1 - \frac{1}{\sqrt{x}} = -1 \Rightarrow \frac{1}{\sqrt{x}} = 2 \quad \text{or } \sqrt{x} = \frac{1}{2} \\ x = \frac{1}{4}$$

$$\therefore P = \left(\frac{1}{4}, -3\frac{3}{4}\right)$$

Question 9

$$(a) (i) t=2, R = \frac{200}{5} + 10 \text{ cm}^3/\text{s} = 50 \text{ cm}^3/\text{s}$$

$$(ii) \frac{dR}{dt} = 100 \left(\frac{1+t^2 - t \cdot 2t}{(1+t^2)^2} \right)$$

$$= 100 \frac{(1-t^2)}{(1+t^2)^2} = 0 \text{ if } t^2=1 \Rightarrow t=1 \text{ since } t \geq 0$$

$$t=0, \frac{dR}{dt} > 0 ; t=2, \frac{dR}{dt} < 0 \Rightarrow \text{max}$$

$$\therefore \text{max } R = \frac{100}{2} + 10 \text{ cm}^3/\text{s} = 60 \text{ cm}^3/\text{s}$$

$$(iii) R = \frac{dV}{dt} \Rightarrow V = \int \frac{100t}{1+t^2} + 10 \, dt$$

$$= 50 \ln(1+t^2) + 10t + c$$

$$: 69 = 0 + 0 + c, c = 69$$

$$\therefore V = 50 \ln(1+t^2) + 10t + 69$$

$$t=10, V = 50 \ln 101 + 100 + 69$$

$$= 399.75 \dots \text{ cm}^3$$

$$= 400 \text{ cm}^3, 3 \text{ sig figs (i.e. nearest cm}^3\text{)}$$

$$(b) (i) V = \pi \int_0^1 x^2 \, dy : y = x^{\frac{3}{2}}$$

$$\therefore (x^{\frac{3}{2}})^{\frac{4}{3}} = x^2 = y^{\frac{4}{3}}$$

$$\therefore V = \pi \int_0^1 y^{\frac{4}{3}} \, dy = \pi \frac{3}{7} [y^{\frac{7}{3}}]_0^1 = \frac{3\pi}{7} \text{ u}^3$$

(ii) (LOTS OF WAYS)

$$\text{e.g. using (i), } V = \frac{3\pi}{7} + \pi \cdot 1^2 \cdot 1 - \frac{3\pi}{7} = \pi \text{ u}^3$$

Question 10

(a) Choose base 10 for obvious reasons (base e okay, of course)

$$\therefore \log_{10} 0.5^n < \log_{10} (12 \times 10^{-2010})$$

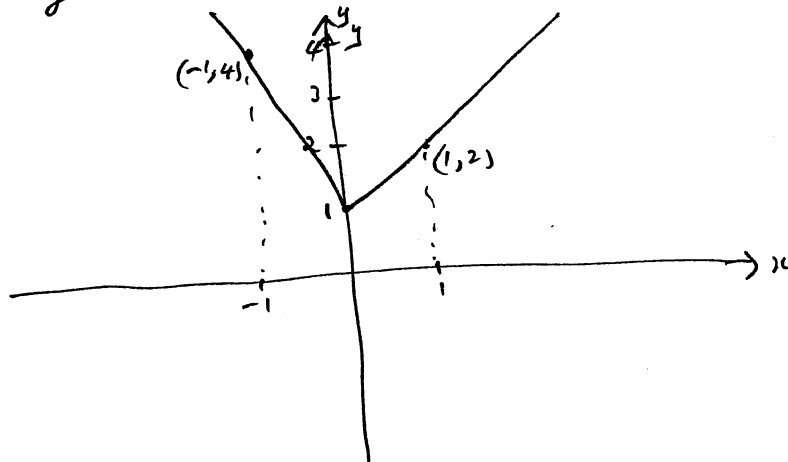
$$\text{or } n \log_{10} 0.5 < \log_{10} 12 - 2010$$

$$\text{or } n > \frac{\log_{10} 12 - 2010}{\log_{10} 0.5} \quad \text{since } \log_{10} 0.5 < 0$$
$$= 6673.49 \dots$$

$$\therefore \text{least } n = 6674$$

(b) (i) $x \geq 0, y = 2x - x + 1 = x + 1$

$x \leq 0, y = -2x - x + 1 = -3x + 1$



(ii) $\therefore I = \frac{1}{2} \cdot (1+4)1 + \frac{1}{2} (1+2)1 = 4$

(D) (c)

(i) ratio intercept theorem in parallel lines (AB, PQ)

$$(ii) \quad \frac{CB}{CQ} = \frac{2CM}{2CN}, \quad (i) \text{ and data}$$
$$= \frac{CM}{CN}$$

(iii) $\angle C$ is common

(ii) $\Rightarrow$ two pairs of corresponding sides are in proportion
with $\angle C$ the included angle
 $\therefore$ similar

(iv) From (iii), $\angle BMC = \angle QNC$, $\left\{ \begin{array}{l} \text{matching} \\ \text{corr. } \angle s \end{array} \right.$ in similar Δs ,
and these angles are corresponding

$\therefore MB \parallel NQ$